

Qatar-10b

R-0042

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-10_260324 · created 2026-07-09T02:19:42+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461123.952 +/- 0.014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.05
Transit depth (Rp/Rs)^2	2.29 +/- 1.5 [%]
Orbital Inclination (inc)	89.32 +/- 2.66
Airmass coefficient 1 (a1)	56.11 +/- 5.13
Airmass coefficient 2 (a2)	-2.99 +/- 1.77
Scatter in the residuals of the lightcurve fit is	9.14 %
Best Comparison Star	#1 - [258, 305]
Optimal Aperture	1.46
Optimal Annulus	3.5
Transit Duration (day)	0.00156 +/- 0.00091

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.151 $\pm$ 0.05 vs 0.1265 (deviation 0.49 σ)
Duration fit vs published	0.00156 d vs 0.1154 d (deviation 125.12 σ)
Mid-time O-C	20.0 min
Depth SNR	3.0
Residual scatter	9.14 %

Light curve

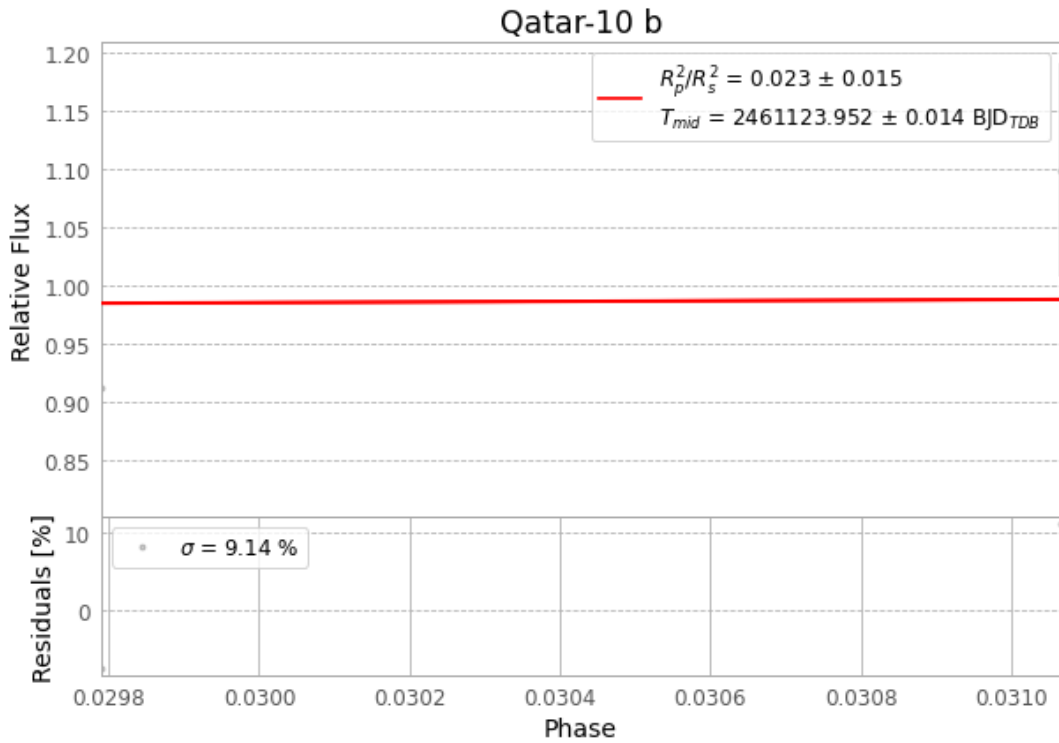


Figure 1 — FinalLightCurve_Qatar-10 b_2026-03-24.png (SHA-256 3fb0b03130df34ac...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [326, 274], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 83a01cfde84fc2ff...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

56 FITS frames (37 MB), Qatar-10260324094515.FITS → Qatar-10260324123015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-10-260324.log (1d97551266c1...), FinalLightCurve_Qatar-10 b_2026-03-24.png (3fb0b03130df...), FinalLightCurve_Qatar-10 b_2026-03-24.csv (3e12fc4d261d...)

Compute resources

Wall time 111.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.